Multimodal Emotion Recognition Using Transformer-Based Fusion on the MELD Dataset

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Abstract

This paper presents a systematic investigation of unimodal and multimodal deep learning approaches for emotion classification, comprising eight model configurations. Unimodal text experiments show that fine-tuned DistilBERT achieves 91.8% accuracy on the Twitter Emotions dataset. Unimodal image experiments on OAHEGA show that ResNet-18 transfer learning (79.6%) substantially outperforms a from-scratch CNN (55.4%). A controlled multimodal experiment demonstrates that dataset alignment is the dominant factor in cross-modal learning: randomly paired unaligned data yields Cohen's Kappa of 0.00, while the synchronized MELD dataset enables genuine learning. Two MELD multimodal variants are evaluated — CNN encoder (42.2%, Kappa 0.29) and ResNet-18 encoder (47.6%, Kappa 0.31) — confirming that transfer learning benefits extend into the multimodal fusion setting.

Introduction

Human emotion plays a central role in communication, decision-making, and social interaction. The ability of computational systems to perceive, interpret, and respond to emotional states – a capability known as Affective Computing – has become increasingly important as AI systems are deployed in contexts requiring empathetic responses. According to a 2023 market analysis by Grand View Research, the global emotion detection and recognition market was valued at approximately USD 47.28 billion and is projected to grow at a compound annual rate of 16.0% through 2030 [1], driven primarily by demand in mental health technology, customer experience analytics, and intelligent tutoring systems. Despite this commercial momentum, the fundamental scientific challenge of reliably classifying human emotion remains unsolved.

The difficulty arises from the inherently multimodal nature of emotional expression. Human beings communicate simultaneously through spoken language, facial muscle movement, vocal prosody, and body posture. Systems that process only one of these channels are technically limited. A facial expression classifier cannot distinguish sarcasm from sincerity; a text classifier cannot detect the forced smile of a person in distress. This limitation is particularly consequential in applications such as mental health monitoring, where misclassifying genuine distress as neutral carries real harm. Ekman and Friesen [2] identified six universal basic emotions – happiness, sadness, anger, fear, disgust, and surprise – that manifest consistently across cultures through facial action units, establishing the scientific basis for automated facial emotion recognition. More recent work has extended this framework to conversational contexts, where emotional state is shaped by dialogue history, speaker relationships, and situational context.

This project investigates the hypothesis that deep learning models trained on aligned multimodal data will outperform unimodal models on emotion classification. We develop a progression of models, unimodal CNN and Transformer-based systems to a full Transformer-CNN multimodal fusion system, evaluating each against common benchmarks. A critical intermediate finding, discovers empirically, is that multimodal fusion on unaligned data is strictly worse than unimodal approaches: not only does it fail to help, but actively degrades performance by introducing contradictory cross-modal signals. This finding motivates the adoption of the MELD [5] dataset and shapes the final system design.

Related Work

For text-based emotion classification, early TF-IDF approaches have been superseded by Transformer architectures [6]. BERT [3] and its distilled variant DistilBERT [4] use self-attention to capture long-range dependencies, achieving state-of-the-art results across NLP benchmarks. For image-based emotion recognition, deep residual networks [7] with skip connections $H(x) = F(x) + x$ enable training of very deep networks and, combined with ImageNet transfer learning, substantially outperform random initialization on small datasets. Multimodal emotion recognition benefits from temporally aligned data: Poria et al. [5] showed MELD enables stronger cross-modal learning than independently collected datasets. Baltrusaitis et al. [8] provide a comprehensive taxonomy of fusion strategies, including the hybrid cross-modal attention approach implemented here.

Problem Statement

We formulate emotion recognition as a supervised multi-class classification problem. Let $x_t \in X_t$ represent the textual input (e.g., a tokenized utterance), and let $x_v \in X_v$ represent the visual input (e.g., a facial image frame). Each input pair is associated with a label $y \in Y$, where Y represents the set of emotion classes.

The objective is to learn a mapping:

$$f(x_t,\, x_v;\, \theta) \rightarrow y$$

where θ represents the parameters of the model.

For unimodal baselines, the problem reduces to:

$$f_t(x_t) \rightarrow y \text{ (text only)}$$

$$f_v(x_v) \rightarrow y \text{ (image only)}$$

In the multimodal setting, the model first encodes each modality into feature representations:

$$h_t = f_t(x_t),\; h_v = f_v(x_v)$$

Optionally, cross-modal attention is applied to allow interaction between modalities before fusion:

$$h'_t = Attn(h_t,\, h_v),\; h'_v = Attn(h_v,\, h_t)$$

The resulting representations are then combined using feature concatenation:

$$h = concat(h'_t,\, h'_v)$$

Finally, the fused representation is passed through a fully connected fusion head. In the implemented model, this fusion head consists of a linear layer, GELU activation, dropout, a second linear layer, another GELU activation and dropout, and a final linear layer that outputs logits for each emotion class:

$$z_1 = GELU(W_1 h + b_1)$$

$$z_2 = GELU(W_2 z_1 + b_2)$$

$$\hat{y} = soft\max(W_3 z_2 + b_3)$$

Where h is the fused representation, W_1, W_2, $and\ W_3$ are learned weight matrices, and $\hat{y}$ is the predicted probability distribution over emotion classes.

The model is trained using cross-entropy loss:

$$L = -\sum w_i \times y_i \times \log(\hat{y_i})$$

where $\hat{y_i}$ is the predicted probability for class i .

In the multimodal setting, the goal is to learn a joint representation that captures complementary information from both modalities to improve classification performance. This is particularly important because emotional expression is often distributed across modalities, and relying on a single modality may result in incomplete or ambiguous predictions.

Datasets

Twitter Emotions Dataset

The Twitter Emotions dataset [9] was used for developing and evaluating unimodal text models. The dataset consists of 416,809 English Twitter messages labeled with one of six emotion categories: sadness (0), joy (1), love (2), anger (3), fear (4), and surprise (5). Each entry contains a raw text field representing the tweet content and an integer label field. The dataset is distributed as a single CSV file (emotions.csv). Text lengths range from a few words to Twitter's character limit, with a median length of approximately 14 tokens after DistilBERT tokenization. The class distribution is moderately imbalanced: joy and sadness together account for approximately 60% of samples, while love and surprise are the least represented categories at roughly 7% and 5% respectively. For all text experiments, an 80/20 stratified train/test split was applied, preserving class proportions across both partitions.

OAHEGA Facial Emotion Recognition Dataset

The OAHEGA dataset [10] was used for developing and evaluating unimodal image models. The dataset contains approximately 15,500 RGB facial images organized into six emotion classes: Ahegao, Angry, Happy, Neutral, Sad, and Surprise. Images were collected from multiple sources including Facebook, Instagram, YouTube video frames, the IMDB celebrity image database, and the AffectNet dataset, producing high intra-class diversity that approximates real-world facial expression variability. All images are provided as pre-cropped face regions. The dataset exhibits significant class imbalance: the Happy, Neutral, and Sad classes each contain approximately 2,600 to 2,800 training samples, while the Ahegao, Angry, and Surprise classes contain only 843, 919, and 864 training samples respectively — a ratio of approximately 3:1 between the most and least represented classes. For all image experiments, a stratified 70/15/15 train/validation/test split was applied, yielding approximately 10,818 training, 2,318 validation, and 2,318 test samples.

Multimodal EmotionLines Dataset (MELD)

The Multimodal EmotionLines Dataset [5] was used for multimodal fusion experiments. MELD is constructed from dialogue clips from the TV series Friends and provides synchronized utterance-level text transcripts, video clips, and audio for 1,433 dialogues comprising 13,708 individual utterances. Each utterance is annotated with one of seven emotion labels: anger, disgust, fear, joy, neutral, sadness, and surprise. The dataset is pre-partitioned into official train, development, and test splits containing 9,989, 1,109, and 2,610 utterances respectively. The class distribution in MELD is severely imbalanced: neutral accounts for 46.8% of all utterances, while disgust and fear together represent only 3.5%. In the test set specifically, neutral has 1,256 samples while disgust and fear have only 68 and 50 samples (imbalance ratios of approximately 25:1 and 18:1 respectively). This extreme imbalance has direct consequences for model evaluation: a model that always predicts neutral would achieve approximately 46% accuracy while learning nothing about emotional content, making accuracy an unreliable standalone metric and necessitating per-class reporting and Cohen's Kappa.

Methodology

A. Data Preprocessing

Text Preprocessing: For the classical ML baseline experiments, raw tweet text was processed through a five-step cleaning pipeline implemented in scikit-learn. First, Unicode characters were normalized to ASCII using the unidecode library to remove diacritics and special symbols that do not contribute to semantic content. Second, all text was converted to lowercase to ensure that capitalization differences (e.g., 'ANGRY' versus 'angry') do not produce distinct vocabulary entries. Third, typographic quotation marks were replaced with standard ASCII equivalents. Fourth, all numeric characters were removed using regex substitution, as numerals rarely carry emotion-bearing information in short social media text. Fifth, multiple consecutive whitespace characters were collapsed to single spaces. The cleaned text was then vectorized using TF-IDF (Term Frequency-Inverse Document Frequency) with English stop-word removal and a maximum vocabulary size of 5,000 features for Random Forest and 3,000 features for the stacking experiment. For the DistilBERT experiments, no manual cleaning was applied. The DistilBERT WordPiece tokenizer handles subword normalization internally, and manual cleaning risks removing signals the pre-trained model has learned to use. Sequences were padded or truncated to a maximum length of 128 tokens with corresponding attention masks.

Image Preprocessing: For the scratch CNN experiments, images were resized to 48×48 pixels using bilinear interpolation to match the network's expected input dimensions. During training, random horizontal flipping (probability 0.5), random rotation (±15 degrees), and color jitter (brightness and contrast variation ±0.2) were applied as data augmentation to improve generalization. For ResNet-18 experiments, images were resized to 224×224 pixels and normalized using ImageNet channel statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Using the correct normalization statistics is critical for transfer learning: the pretrained convolutional filters were optimized assuming this specific input distribution, and applying incorrect normalization shifts the feature activations away from their trained operating range. Validation and test sets received only resize and normalize operations, with no augmentation applied.

MELD Preprocessing: MELD video preprocessing was handled by the prepare_meld.py script, which used FFmpeg to extract a single representative frame (frame index 0) from each utterance video clip. Frames were saved as JPEG images organized by split (train/dev/test) and indexed by dialogue ID and utterance ID. Each frame was then matched to its corresponding row in the annotation CSV files (train_sent_emo.csv, dev_sent_emo.csv, test_sent_emo.csv) by matching dialogue_id and utterance_id fields. Utterances with missing or corrupted frames were excluded. Emotion label strings were canonicalized through an alias dictionary to handle MELD's inconsistent labeling conventions (e.g., 'angry' and 'anger' both map to the canonical label 'anger'). Class weights for the weighted loss function were computed as the inverse frequency of each class in the training split, with label smoothing of 0.02 applied to reduce overconfidence in predictions.

B. Addressing Class Imbalance

Class imbalance was a persistent challenge across all three datasets and required careful handling to avoid models that achieve high accuracy by ignoring minority classes entirely. The weighted cross-entropy loss function assigns a scalar weight w_k to each class k, scaling the contribution of each sample's loss contribution The initial approach used raw inverse class frequency: w_k = N / (K · n_k), where N is the total number of training samples, K is the number of classes, and n_k is the number of samples in class k combined with WeightedRandomSampler produced catastrophic over-correction in the OAHEGA image experiments: test accuracy dropped from 55% to 20% because both mechanisms compounded, causing the model to predict only minority classes for all inputs. This failure motivated a two-part design change: WeightedRandomSampler was removed entirely, and loss weights were softened by replacing raw inverse frequency with square-root inverse frequency $w_k = 1/\sqrt{n_k}$, normalized to sum to K. The square-root transformation compresses the weight range while preserving relative class priorities, producing a gentler correction. For the MELD multimodal models, train.py computes raw inverse frequency weights (w_k= N / n_k) with label smoothing of 0.02, as the more extreme MELD imbalance ratios (up to 37×) were found empirically to require stronger correction than the OAHEGA case.

C. Models 1 and 2: Classical ML Baselines (Text)

Two classical ML baselines were developed to establish a performance floor for the text modality. Random Forest and Logistic Regression were chosen because they are well-understood, fast to train, and provide a meaningful comparison point against deep learning approaches. Both use TF-IDF features with English stop-word removal. Random Forest used 100 trees with Gini impurity on 5,000 features. Logistic Regression used L2 regularization with the lbfgs solver on 3,000 features.

D. Model 3: DistilBERT Transformer (Text)

The primary deep learning text model fine-tunes a pretrained DistilBERT for 6-class classification. DistilBERT was selected as the primary text model because it retains the contextual understanding of BERT while being significantly smaller and faster to fine-tune, making it practical for this project's compute constraints. Fine-tuning used AdamW at lr=2e-5 with batch size 128 for 10 epochs on a 10,000-sample subset, with a full dataset run planned on the HPC. The DistilBERT text encoders pretrained CLS representations are also used directly as the text branch of both MELD multimodal models.

E. Model 4: Custom CNN from Scratch (Image)

The first image model is a custom CNN built from random initialization as a baseline image model to measure how much benefit transfer learning provides. Three convolutional blocks with filter counts 32 → 64 → 128, each containing Conv2d, BatchNorm, ReLU, MaxPool, and Dropout2d, feed into two fully connected layers. The Adam optimizer (lr=0.001) with StepLR (step=5, gamma=0.5) trained the model for 30 epochs.

F. Model 5: ResNet-18 Transfer Learning (Image, OAHEGA)

The scratch CNN's limitations motivated ResNet-18 transfer learning on OAHEGA. ResNet-18 was chosen because ImageNet pretraining gives the model a strong foundation of visual features — edges, textures, and shapes — that transfer effectively to facial expression recognition. This is especially valuable when labeled training data is limited, as is the case with OAHEGA. The final fully connected layer (512 → 1000 classes) was replaced with a custom head: Dropout(0.4) →

Linear(512,256) → ReLU → Dropout(0.3) → Linear(256,6). Training used a two-phase strategy: Phase 1 froze the backbone and trained only the custom classification head to avoid catastrophic forgetting. Phase 2 unfroze layer4 for end-to-end fine-tuning with differential learning rates of 1e-4 for the backbone and 1e-3 for the head. The best checkpoint was selected by minimum validation loss at Phase 2 epoch 5 (val loss 0.557).

G. Model 6: Multimodal Fusion – Unaligned Data (Failed Baseline)

Before adopting MELD, a multimodal experiment paired the Twitter and OAHEGA datasets by matching emotion labels. This approach was theoretically motivated as a way to create paired training data without a purpose-built multimodal dataset. However, the paired samples share only a label — the facial expression and the tweet come from entirely different people and contexts, meaning the image provides no genuine additional signal about the text's emotional content. This established the alignment requirement that motivated MELD adoption.

H. Model 7: Multimodal Fusion with CNN Encoder (MELD)

The first MELD multimodal model uses a from-scratch CNN image encoder paired with a DistilBERT text encoder and cross-modal attention fusion. The architecture consists of: (1) a TransformerTextEncoder that loads pretrained DistilBERT, extracts the CLS token from the final hidden state (dimension 768), and projects it through Linear(768,256) → GELU → Dropout(0.2) → LayerNorm(256) to produce a 256-dimensional text feature vector; (2) an ImageCNNEncoder with three double-convolution blocks (32, 64, 128 filters), global average pooling, and Linear(128,256) → ReLU → LayerNorm(256) projection; (3) CrossModalAttention modules applying scaled dot-product attention between the two feature vectors, with projection dimensions of 128; and (4) a fusion head processing the concatenated 512-dimensional vector through Linear(512,256) → GELU → Dropout(0.3) → Linear(256,128) → GELU → Dropout(0.3) → Linear(128,7). Total trainable parameters: 67,309,031. Training used AdamW (lr=2e-5, weight_decay=1e-4), cosine annealing (T_max=10, eta_min=1e-6), gradient clipping (max_norm=1.0), and early stopping (patience=4).

Model 8: Final System Architecture

The final multimodal system (Model 8) consists of three jointly trained components: a Transformer text encoder, a ResNet-18 image encoder, and a cross-modal attention fusion head.

The text encoder loads pretrained DistilBERT, extracts the CLS token from the final hidden state (dim 768), and projects through Linear(768,256) → GELU Dropout(0.2) → LayerNorm(256). The image encoder loads pretrained ResNet-18 with all layers frozen except layer4, replaces the original FC layer with Identity(), and projects through Linear(512,256) → ReLU Dropout(0.3) → LayerNorm(256). Both encoders produce 256-dimensional feature vectors. CrossModalAttention computes scaled dot-product attention with projection dimension 128: image features attend to text features and vice versa, allowing each modality to emphasize regions informative to the other. The attended features are concatenated (512-dim) and passed through the fusion head: Linear(512,256) → GELU Dropout(0.3) → Linear(256,128) GELU Dropout(0.3) → Linear(128,7). Total trainable parameters: 75,513,607.

Results

The models were evaluated to answer three main questions: whether text or image features were more reliable for emotion recognition, whether pretrained models improved performance over models from scratch, and whether multimodal fusion helped when text and image inputs were properly aligned. Therefore, the results are organized around unimodal text performance, unimodal image perfromance, and multimodal fusion performance rather than accuracy alone. This structure allows each result to show not only which model performed best, but also why certain modeling choices succeeded or failed.

A. Overall Performance Comparison

Model	Accuracy	Macro F1	Kappa	ROC-AUC
Random Forest (TF-IDF)	85.6%	—	—	—
Logistic Regression (TF-IDF)	89.7%	—	—	—
DistilBERT (fine-tuned)	91.8%	—	—	—
CNN from scratch	55.4%	0.41	—	—

ResNet-18 (transfer learning)	79.6%	0.80	—	—
Multimodal — unaligned (failed)	38.5%	0.14	0.00	0.50
Multimodal — MELD + CNN	42.2%	0.31	0.29	0.74
Multimodal — MELD + ResNet-18	47.6%	0.31	0.31	0.74

Complete Model Performance Summary

Main takeaway: The strongest overall model was the fine-tuned DistilBERT text model, which reached 91.8% accuracy. This shows that, for this project, text was the most reliable single modality for emotion classification. However, the image results also show that architecture mattered greatly: ResNet-18 improved image accuracy from 55.4% to 79.6% compared with the from scratch CNN. The multimodal models did not outperform the best unimodal text model, but the aligned MELD models performed meaningfully better than the unaligned multimodal model, showing that multimodal fusion only works when the text and visual inputs refer to the same context.

B. Text Modality

The text modality was the strongest individual signal for emotion recognition in this project. Logistic Regression (89.7%) outperforms Random Forest (85.6%) despite fewer TF-IDF features, consistent with high-dimensional sparse representations favoring linear decision boundaries. DistilBERT outperforms Logistic Regression by 2.1 points because it understands word context rather than just counting words. A phrase like "I'm fine" and "that's a fine example" look identical to TF-IDF but produce very different representations in DistilBERT. These same pretrained representations are used as the text encoder in both MELD multimodal models.

C. Image Modality

Class	CNN Prec.	CNN Rec.	CNN F1	ResNet Prec.	ResNet Rec.	ResNet F1
Ahegao	0.79	0.85	0.82	0.94	0.96	0.95
Angry	0.00	0.00	0.00	0.77	0.63	0.69

Happy	0.78	0.87	0.82	0.91	0.92	0.91
Neutral	0.50	0.63	0.56	0.75	0.71	0.73
Sad	0.34	0.33	0.34	0.72	0.79	0.75
Surprise	0.41	0.38	0.39	0.73	0.75	0.74
Macro Avg	0.47	0.51	0.49	0.80	0.79	0.80

Per-Class Results — Image Modality (OAHEGA Test Set)

The image results show that transfer learning was necessary for stringer facial emotion recognition. The scratch CNN's complete failure on Angry (0% F1) is not a simple imbalance problem — Ahegao has fewer training samples (843 vs 919) yet achieves 82% F1. This suggests that the scratch CNN did not learn strong enough visual features to separate visually similar emotions:

Angry and Sad share visual overlap (downturned features, furrowed brows) requiring fine-grained detectors that random initialization cannot learn from ~900 samples. ResNet-18's pretrained feature hierarchy provides this capability: Angry improves from 0% to 69% F1 while every other class simultaneously improves, confirming the benefit is architectural rather than data-driven.

D. Multimodal Analysis

Metric	Unaligned	MELD + CNN	MELD + ResNet-18
Accuracy	38.5%	42.2%	47.6%
Top-2 Accuracy	~75%	72.6%	66.8%
Macro F1	0.14	0.31	0.31
F1 (weighted)	0.21	0.45	0.49
Cohen's Kappa	0.00	0.29	0.31
ROC-AUC	0.50	0.74	0.74
MCC	~0.00	0.32	0.33

Multimodal Model Comparison

Training and validation loss/accuracy curves for Model 8 (MELD + ResNet-18) over 12 epochs. Train and validation loss converge through epoch 6 before diverging; early stopping selected the epoch 8 checkpoint.

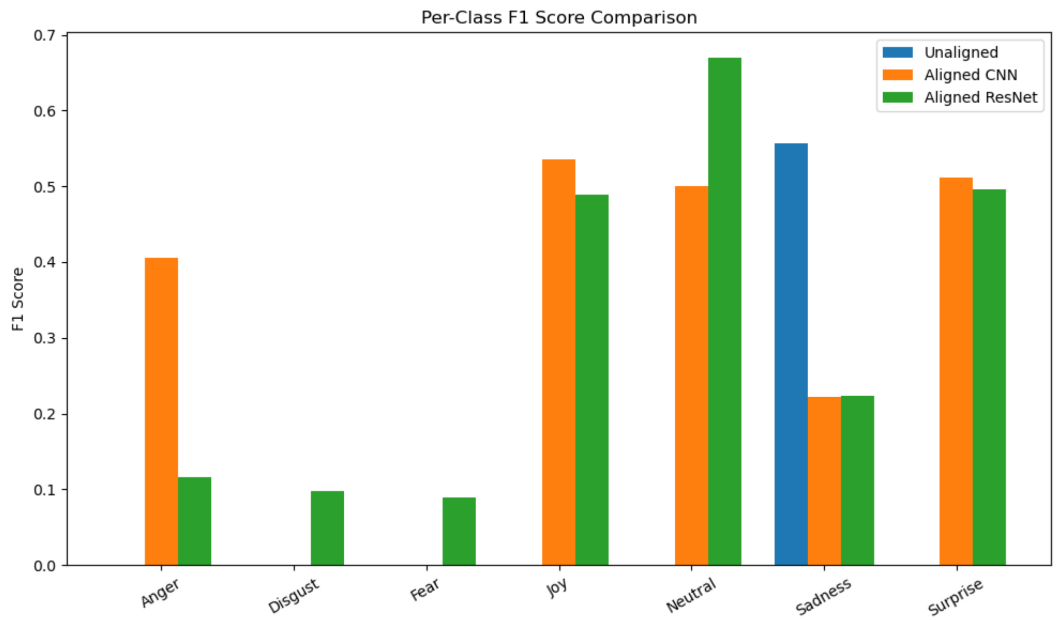
The unaligned model's Kappa of 0.00 and ROC-AUC of 0.50 confirm predictions statistically indistinguishable from random guessing, despite 38.5% accuracy, illustrating why accuracy alone is misleading on imbalanced data, and multimodal fusion does not work when the text and image inputs are not meaningfully aligned.

The MELD CNN model's Kappa of 0.29 indicates genuine learning, and the ResNet-18 variant further improves Kappa 0.31 showing that unaligned text-image pairs allowed the model to learn real multimodal patterns instead of guessing.

Replacing the CNN encoder with ResNet-18 yields: accuracy +5.4 points, weighted F1 +0.04, and disgust recall improving from 0.00 to 0.38. This means the ResNet-based model can now detect a Class that the CNN-based model missed completely. Neutral F1 improves from 0.50 to 0.67. However, anger recall decreases from 0.46 to 0.07, with 35% of anger samples misclassified as joy. This tradeoff reveals that improving the image encoder shifts fusion weight toward visual-dominant predictions, benefiting visually distinct emotions (disgust, surprise) while hurting emotions better resolved by text (anger). This motivates modality gating as future work.

Class-Level Performance Analysis

To further evaluate model performance across individual emotion classes, per-class F1 scores were computed for the multimodal models.



Per-class F1 score comparison for unaligned, aligned CNN, and aligned ResNet multimodal models.

The unaligned multimodal model shows non-zero performance only for the sadness class, with all other classes receiving an F1 score of 0, indicating the model predicts a single dominant class across most inputs. In contrast, the aligned multimodal model using a CNN encoder demonstrates non-zero F1 scores across several classes including anger, joy, neutral, sadness, and surprise, though performance remains uneven with disgust and fear receiving no correct predictions. The ResNet-based multimodal model further expands coverage across classes, achieving non-zero F1 scores for all emotion categories. Compared to the CNN-based model, it shows improved balance across classes, although performance still varies significantly between categories.

Conclusion

This project investigated unimodal and multimodal approaches to emotion recognition using text and image data. Across all experiments, Transformer-based text models achieved the highest performance, with DistilBERT outperforming both classical machine learning baselines and image-based models. Image models showed strong improvements when using pretrained architectures, as demonstrated by the performance gain from a CNN trained from scratch to a ResNet-18 model.

A central finding of this work is the importance of dataset alignment in multimodal learning. The multimodal model trained on unaligned data failed to learn meaningful relationships between modalities, while models trained on the aligned MELD dataset showed clear improvements in both accuracy and class-level performance. These results highlight that multimodal systems require carefully synchronized inputs to be effective.

Despite these improvements, the multimodal models did not surpass the performance of the best unimodal text model. This indicates that simple combining modalities is not sufficient to achieve better results and that more advanced fusion strategies may be required to fully leverage multimodal information.

This work is subject to several limitations. The multimodal models relied on a single frame representation of video data, which may not fully capture dynamics of facial expressions. Additionally, class imbalance in the datasets posed a significant challenge. Although weighting strategies were applied during training to alleviate imbalance, minority classes such as disgust and fear remained difficult to classify accurately, indicating that further improvements in imbalance handling are needed.

Potential future work could explore more sophisticated multimodal fusion techniques, including attention-based fusion and temporal modeling of video sequences. Incorporating additional modalities such as audio, as well as applying more advanced methods for handling class imbalance, may further improve performance. Expanding evaluation to more diverse datasets would also provide a more comprehensive understanding of model generalization.

References

[1] Grand View Research, "Emotion Detection and Recognition Market Report," 2023. [Online]. Available: https://www.grandviewresearch.com/industry-analysis/emotion-detection-recognition-market-report

[2] P. Ekman and W. V. Friesen, *Facial Action Coding System.* Consulting Psychologists Press, 1978.

[3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.

[5] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "MELD: A multimodal multi-party dataset for emotion recognition in conversations," in *Proc. ACL*, 2019, pp. 527–536.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. NeurIPS*, 2017.

[7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. CVPR*, 2016, pp. 770–778.

[8] T. Baltrusaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 423–443, 2018.

[9] B. Jikadara, "Emotions Dataset," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/bhavikjikadara/emotions-dataset

[10] V. Kovenko and V. Shevchuk, "OAHEGA: Emotion Recognition Dataset," Mendeley Data, V2, 2021. doi: 10.17632/5ck5zz6f2c.2